EGFR fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes EGFR gene fusions from the msk_impact_50k_2026 study, covering 55 fusion events. 28/55 fusions (50.9%) were in-frame. 41/55 fusions (74.5%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test (p=0.0114539, statistically significant at alpha=0.05). No literature benchmark is configured for EGFR.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 55 analyzed fusion events, identifying 31 distinct fusion partner genes. The most frequent partner was SEPTIN14, observed in 18 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 25/28 (89.3%) of in-frame fusions retained the domain, p=0.0114539 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.227723.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 6/28 (21.4%) of in-frame fusions disrupted the domain, p=0.991541 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.910891.

Cutpoint detection

Cutpoint detection scanned 55 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 981 aa (permutation-corrected p=0.128713, not statistically significant at alpha=0.05). This is 16 aa from the nearest configured domain boundary at 965 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=41) and "not_retained" (n=14). For Frame_status, retained: 25/31 (80.6%, 95% CI 63.7%-90.8%); not_retained: 3/9 (33.3%, 95% CI 12.1%-64.6%). Welch's t-test comparing Tumor_variant_count between groups gave means of 358.805 vs 214.071, p=0.389355 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (31 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: EGFR has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SEPTIN14	18	0.32727272727272727	0.06425935377650496	0.004053354643708729	0.9779763991120458	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3219464603947537	1
RAD51	2	0.03636363636363636	0.06425935377650496	0.004053354643708729	0.9989484752891693	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23781954454859494	2
LANCL2	3	0.05454545454545454	0.06425935377650496	0.004053354643708729	0.961794602173151	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23770100903573768	3
ELAPOR2	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	1.0	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2325227278006741	4
PKD1L1	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.9989484752891693	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23236499909404948	5
VPS41	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.9947423764458465	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23173408426755107	6
ZNF713	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.9400630914826499	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22353219152307158	7
VWC2	2	0.03636363636363636	0.06425935377650496	0.004053354643708729	0.9032597266035752	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22346623224575585	8
NIPSNAP2	3	0.05454545454545454	0.06425935377650496	0.004053354643708729	0.8647038205397827	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22313739179073241	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TNRC18	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.9232386961093586	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2210085322170779	10
BLTP3B	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.9011566771819137	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21769622937796115	11
CDH7	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.8338590956887487	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2076015921539864	12
FADD	2	0.03636363636363636	0.06425935377650496	0.004053354643708729	0.6908517350157728	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19160503350758548	13
CADPS	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.6919032597266035	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18630821675966464	14
EEA1	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.6919032597266035	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18630821675966464	15
KIF5B	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.6919032597266035	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18630821675966464	16
NUMA1	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.6919032597266035	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18630821675966464	17
VSTM2A	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.6908517350157728	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18615048805304002	18
PDE1C	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.6529968454258674	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18047225461455424	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
YIF1B	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.6414300736067298	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1787372388416835 8	20
TUT7	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.6277602523659306	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1766867656555637	21
CSF2RA	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.3217665615141956	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1307877120278034 5	22
DDC	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.28075709779179814	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1246362924694438 3	23
GARS1	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.28075709779179814	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1246362924694438 3	24
SEL1L	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.28075709779179814	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1246362924694438 3	25
TNS3	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.28075709779179814	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1246362924694438 3	26
PCDH15	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.27129337539432175	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1232167341098223 7	27
VOPP1	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.2302839116719243	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1170653145514627 7	28
SCAF4	1	0.0181818181818181 8	0.06425935377650496	0.004053354643708729	0.11777076761303895	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1001883429426299 6	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ZPBP	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.006309148264984188	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08346910004042174	30
PLOD3	1	0.01818181818181818	0.06425935377650496	0.004053354643708729	0.0	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08252272780067411	31

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
30	1	40	0	14	None	1.0	-0.0
36	2	39	0	14	None	1.0	-0.0
142	3	38	0	14	None	0.5624166190203925	0.24994185469837954
249	3	38	1	13	1.0263157894736843	1.0	-0.0
288	3	38	2	12	0.47368421052631576	0.5924468510484049	0.22735060452753406
297	7	34	2	12	1.2352941176470589	1.0	-0.0
336	8	33	2	12	1.4545454545454546	1.0	-0.0
599	9	32	2	12	1.6875	0.7093311767774617	0.14915095146564436
627	9	32	3	11	1.03125	1.0	-0.0
640	10	31	3	11	1.1827956989247312	1.0	-0.0
651	11	30	3	11	1.3444444444444446	1.0	-0.0
687	11	30	6	8	0.4888888888888889	0.3219137899102455	0.49226041892280514
688	15	26	6	8	0.7692307692307693	0.7548366596998944	0.12214701587495107
823	15	26	7	7	0.5769230769230769	0.528760430668828	0.27674105235972185
875	15	26	8	6	0.4326923076923077	0.21856306955359578	0.6604232186170306
876	15	26	9	5	0.32051282051282054	0.1175979559898796	0.9296002268145651
981	15	26	10	4	0.23076923076923078	0.031971018585759146	1.4952435270336717
982	31	10	10	4	1.24	0.7358375514955705	0.1332180528854592
983	31	10	13	1	0.23846153846153847	0.25480492869908067	0.5937921756847577
986	32	9	13	1	0.27350427350427353	0.4229368950729017	0.37372442734387856
994	33	8	13	1	0.3173076923076923	0.4208384837822578	0.3758845527657329
1003	34	7	13	1	0.37362637362637363	0.663943317750199	0.17786899569082928

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1038	35	6	13	1	0.44871794871794873	0.663943317750199	0.17786899569082928
1054	38	3	13	1	0.9743589743589743	1.0	-0.0
1055	38	3	14	0	0.0	0.5624166190203926	0.24994185469837946
1060	39	2	14	0	0.0	1.0	-0.0
1075	40	1	14	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

6	13
22	14

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf01030	Receptor L domain	55	36	1	18	0.012328117591275486
auto_disruption_pf00757	Furin-like cysteine rich region	55	35	7	13	0.1103896103896104
auto_disruption_pf14843	Growth factor receptor domain IV	55	39	2	14	0.07623106060606061

domain_retention tables

frame_domain_contingency_table

25	16
3	11

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	55	41	3	11	0.10276679841897234

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
18	43	55	0.7818181818181819

frequency tables

Partner_gene_counts

Partner_gene	Event_count
BLTP3B	1
CADPS	1
CDH7	1
CSF2RA	1
DDC	1
EEA1	1
ELAPOR2	1
FADD	2
GARS1	1
KIF5B	1
LANCL2	3
NIPSNAP2	3
NUMA1	1
PCDH15	1
PDE1C	1
PKD1L1	1
PLOD3	1
RAD51	2
SCAF4	1
SEL1L	1
SEPTIN14	18
TNRC18	1
TNS3	1
TUT7	1
VOPP1	1
VPS41	1
VSTM2A	1
VWC2	2
YIF1B	1

Partner_gene	Event_count
ZNF713	1
ZPBP	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031404-T01-IM6-2	P-0031404-T01-IM6		BLTP3B-EGFR	BLTP3B	unknown	five_prime	55270272	27	1075	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - UHRF1BP1L (NM_015054) Rearrangement :t(7;12)(p11.2;q23.1)(chr7:g.55270272::chr12:g.100460508) | Protein Fusion: mid-exon {EGFR:UHRF1BP1L}	NA
EVT-P-0050679-T02-IM6-3	P-0050679-T02-IM6		CADPS-EGFR	CADPS	out-of-frame	three_prime	55241099	18	688	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		CADPS (NM_003716) - EGFR (NM_005228) fusion: t(3;7)(p14.2;p11.2)(chr3:g.62537410::chr7:g.55241099) | Protein Fusion: out of frame {CADPS:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038683-T01-IM6-4	P-0038683-T01-IM6		CSF2RA-EGFR	CSF2RA	unknown	three_prime	55224220	9	336	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	CSF2RA (NM_001161530) - EGFR (NM_005228) rearrangement: t(7;X)(p11.2;p22.33)(chr7:g.55224220::chrX:g.1401662) | Protein Fusion: mid-exon {CSF2RA:EGFR}	NA
EVT-P-0016795-T01-IM6-5	P-0016795-T01-IM6		DDC-EGFR	DDC	out-of-frame	three_prime	55222204	8	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	DDC (NM_000790) - EGFR (NM_005228) Rearrangement : c.435+4059:DDC_c.889+359:EGFRinv | Protein Fusion: out of frame {DDC:EGFR}	NA
EVT-P-0018389-T02-IM6-6	P-0018389-T02-IM6		EEA1-EGFR	EEA1	in-frame	three_prime	55241575	18	688	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EEA1 (NM_003566) - EGFR (NM_005228) rearrangement: t(7;12)(p11.2;q22)(chr7:g.55241575::chr12:g.932211089) | Protein Fusion: in frame {EEA1:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064660-T01-IM7-9	P-0064660-T01-IM7		EGFR-CDH7	CDH7	out-of-frame	five_prime	55254206	20	823	True	disrupted	0.43873517786561267	True	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - CDH7 (NM_033646) rearrangement: t(7;18)(p11.2;q22.1)(chr7:g.55254206::chr18:g.63486938) | Protein Fusion: out of frame {EGFR:CDH7}	NA
EVT-P-0034892-T01-IM6-497	P-0034892-T01-IM6		EGFR-ELAPOR2	ELAPOR2	unknown	three_prime	55268103	24	981	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EGFR (NM_005228) rearrangement: c.2943:EGFR_chr7:g.86587163del | Antisense Fusion	NA
EVT-P-0059222-T03-IM7-500	P-0059222-T03-IM7		EGFR-LANCL2	LANCL2	in-frame	five_prime	55259704	21	875	True	disrupted	0.6442687747035574	True	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - LANCL2 (NM_018697) rearrangement: c.2625+137:EGFR_c.1008+3812:LANCL2del | Protein Fusion: in frame {EGFR:LANCL2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057880-T01-IM6-501	P-0057880-T01-IM6		EGFR-LANCL2	LANCL2	in-frame	five_prime	55268871	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - LANCL2 (NM_018697) fusion: c.2947-10:EGFR_c.205-8690:LANCL2del | Protein Fusion: in frame {EGFR:LANCL2}	NA
EVT-P-0063142-T01-IM7-503	P-0063142-T01-IM7		EGFR-PCDH15	PCDH15	unknown	five_prime	55221820	7	288	False	lost	0.0	False	Receptor L domain	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Furin-like cysteine rich region	EGFR (NM_005228) - PCDH15 (NM_001142763) rearrangement: t(7;10)(p11.2;q21.1)(chr7:g.55221820::chr10:g.56187463) | Protein Fusion: mid-exon {EGFR:PCDH15}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042236-T01-IM6-504	P-0042236-T01-IM6		EGFR-PKD1L1	PKD1L1	in-frame	five_prime	55268824	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - PKD1L1 (NM_138295) fusion: c.2947-57:EGFR_c.8527-7825:PKD1L1inv | Protein Fusion: in frame {EGFR:PKD1L1}	NA
EVT-P-0041149-T04-IM6-505	P-0041149-T04-IM6		EGFR-RAD51	RAD51	in-frame	five_prime	55268799	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189) | Protein Fusion: in frame {EGFR:RAD51}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041149-T01-IM6-506	P-0041149-T01-IM6		EGFR-RAD51	RAD51	in-frame	five_prime	55268799	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189) | Protein Fusion: in frame {EGFR:RAD51}	NA
EVT-P-0034594-T01-IM6-507	P-0034594-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268544	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-110:EGFR_c.1120-1204:SEPT14 | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038138-T01-IM6-508	P-0038138-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268381	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+275:EGFR_c.1119+2043:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0037526-T01-IM6-509	P-0037526-T01-IM6		EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268916	25	994	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2982:EGFR_c.1119+1151:SEPT14inv | Protein Fusion: mid-exon {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039255-T02-IM6-510	P-0039255-T02-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55270807	27	1091	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3271+489:EGFR_c.818-3507:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0032604-T01-IM6-512	P-0032604-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268771	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110:EGFR_c.1120-1204:SEPT14 | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031753-T01-IM6-513	P-0031753-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268412	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+306:EGFR_c.1119+2531:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0005701-T01-IM5-514	P-0005701-T01-IM5		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269490	26	1054	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14inv | Protein fusion: in frame (EGFR-SEPT14)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043525-T01-IM6-515	P-0043525-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268859	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0029159-T01-IM6-516	P-0029159-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	three_prime	55269850	27	1055	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		SEPT14 (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv | Protein Fusion: in frame {SEPT14:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028241-T01-IM6-517	P-0028241-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268525	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2947-356:EGFR_c.1119+4312:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0025191-T02-IM6-518	P-0025191-T02-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269542	26	1054	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3162+67:EGFR_c.1120-3396:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0061245-T01-IM7-519	P-0061245-T01-IM7		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268815	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement: c.2947-66:EGFR_c.1119+2973:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0014220-T02-IM6-520	P-0014220-T02-IM6		EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268942	25	1003	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-25 fused with SEPT14 exon 10): c.3008:EGFR_c.1120-4102:SEPT14inv | Protein Fusion: mid-exon {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023043-T01-IM6-521	P-0023043-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268179	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion (EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+73:EGFR_c.1120-1180:SEPT14 | Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0019603-T01-IM6-522	P-0019603-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268171	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion ((EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+65:EGFR_c.1119+3306:SEPT14 | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047356-T01-IM6-523	P-0047356-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268316	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2946+210:EGFR_c.1120-2817:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0016256-T01-IM6-524	P-0016256-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268390	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion(EGFR exons 1-24 fused to SEPT14 exons 9-10): c.2946+284:EGFR_c.1120-3940:SEPT14inv | Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015179-T01-IM6-525	P-0015179-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268650	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_002529) - SEPT14 (NM_014857) fusion (EGFR exons 1-24 fused with SEPT14 exon 10): c.2947-231:EGFR_c.1119+3910:SEPT14inv | Protein fusion: in frame (EGFR-SEPT14)	NA
EVT-P-0058375-T01-IM6-526	P-0058375-T01-IM6		EGFR-TNRC18	TNRC18	out-of-frame	five_prime	55269897	26	1054	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - TNRC18 (NM_001080495) fusion: c.3163-313:EGFR_c.5720-6144:TNRC18inv | Protein Fusion: out of frame {EGFR:TNRC18}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016009-T01-IM6-527	P-0016009-T01-IM6		EGFR-VOPP1	VOPP1	in-frame	five_prime	55220393	6	249	True	lost	0.0	False	Receptor L domain	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Furin-like cysteine rich region	EGFR (NM_005228) - VOPP1 (NM_030796): c.747+36:EGFR_c.55-23161VOPP1inv | Protein Fusion: in frame {EGFR:VOPP1}	NA
EVT-P-0039262-T01-IM6-529	P-0039262-T01-IM6		EGFR-VSTM2A	VSTM2A	out-of-frame	five_prime	55241509	17	687	True	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - VSTM2A (NM_182546) rearrangement: c.2062-105:EGFR_c.634+2190:VSTM2A dup | Protein Fusion: out of frame {EGFR:VSTM2A}	NA
EVT-P-0042236-T01-IM6-530	P-0042236-T01-IM6		EGFR-VWC2	VWC2	unknown	five_prime	55270225	27	1060	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - VWC2 (NM_198570) fusion: c.3178:EGFR_c.827-13914:VWC2dup | Protein Fusion: mid-exon {EGFR:VWC2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043097-T01-IM6-531	P-0043097-T01-IM6		EGFR-ZNF713	ZNF713	in-frame	five_prime	55269400	25	1038	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - ZNF713 (NM_182633) fusion: c.3115-28:EGFR_c.49-962:ZNF713del | Protein Fusion: in frame {EGFR:ZNF713}	NA
EVT-P-0027944-T01-IM6-534	P-0027944-T01-IM6		FADD-EGFR	FADD	unknown	five_prime	55241077	17	687	True	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) | Protein Fusion: mid-exon {EGFR:FADD}	NA
EVT-P-0027944-T03-IM6-535	P-0027944-T03-IM6		FADD-EGFR	FADD	unknown	five_prime	55241077	17	687	True	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) | Protein Fusion: mid-exon {EGFR:FADD}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0063743-T01-IM7-536	P-0063743-T01-IM7		GARS1-EGFR	GARS1	out-of-frame	three_prime	55222368	8	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	GARS (NM_002047) - EGFR (NM_005228) fusion: c.1032-338:GARS_c.889+523:EGFRdel | Protein Fusion: out of frame {GARS:EGFR}	NA
EVT-P-0021038-T01-IM6-538	P-0021038-T01-IM6		KIF5B-EGFR	KIF5B	in-frame	three_prime	55241486	18	688	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		KIF5B (NM_004521) - EGFR (NM_005228) rearrangement: t(7;10)(p11.2;q11.22)(chr7:g.55241486::chr10:g.32312089) | Protein Fusion: in frame {KIF5B:EGFR}	NA
EVT-P-0025018-T01-IM6-539	P-0025018-T01-IM6		LANCL2-EGFR	LANCL2	out-of-frame	three_prime	55268537	25	983	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		LANCL2 (NM_018697) - EGFR (NM_005228) rearrangement: c.1259-255:LANCL2_c.2947-344:EGFRdup | Protein Fusion: out of frame {LANCL2:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059222-T01-IM7-540	P-0059222-T01-IM7		NIPSNAP2-EGFR	NIPSNAP2	unknown	three_prime	55233046	15	599	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region	Growth factor receptor domain IV	GBAS (NM_001483) - EGFR (NM_005228) fusion: c.92+717:GBAS_c.1796:EGFRdup | Protein Fusion: mid-exon {GBAS:EGFR}	NA
EVT-P-0037759-T01-IM6-541	P-0037759-T01-IM6		NIPSNAP2-EGFR	NIPSNAP2	out-of-frame	three_prime	55268444	25	983	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		GBAS (NM_001483) - EGFR(NM_005228) Rearrangement: c.2946+338:EGFR_c.279-1159:GBASdup | Protein Fusion: out of frame {GBAS:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037759-T03-IM6-542	P-0037759-T03-IM6		NIPSNAP2-EGFR	NIPSNAP2	out-of-frame	three_prime	55268444	25	983	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		GBAS (NM_001483) - EGFR (NM_005228) rearrangement: c.279-1159:GBAS_c.2946+338:EGFRdup | Protein Fusion: out of frame {GBAS:EGFR}	NA
EVT-P-0036028-T01-IM6-543	P-0036028-T01-IM6		NUMA1-EGFR	NUMA1	in-frame	three_prime	55241128	18	688	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		NUMA1 (NM_006185) - EGFR (NM_005228) rearrangement: t(7;11)(p11.2;q13.4)(chr7:g.55241128::chr11:g.71728415) | Protein Fusion: in frame {NUMA1:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065470-T01-IM7-545	P-0065470-T01-IM7		PDE1C-EGFR	PDE1C	unknown	three_prime	55240707	17	651	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		PDE1C (NM_001191058) - EGFR (NM_005228) fusion: c.86-15224:PDE1C_c.1951:EGFRinv | Protein Fusion: mid-exon {PDE1C:EGFR}	NA
EVT-P-0059032-T01-IM7-546	P-0059032-T01-IM7		PLOD3-EGFR	PLOD3	out-of-frame	three_prime	55179084	2	30	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			PLOD3 (NM_001084) - EGFR (NM_005228) fusion: c.2062-9:PLOD3_c.89-30895:EGFRinv | Protein Fusion: out of frame {PLOD3:EGFR}	NA
EVT-P-0028484-T02-IM6-548	P-0028484-T02-IM6		SCAF4-EGFR	SCAF4	out-of-frame	three_prime	55214145	4	142	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Furin-like cysteine rich region; Growth factor receptor domain IV		Receptor L domain	SCAF4 (NM_020706) - EGFR (NM_005228) fusion: t(7;21)(p11.2;q22.11)(chr7:g.55214145::chr21:g.33094864) | Protein Fusion: out of frame {SCAF4:EGFR}	NA

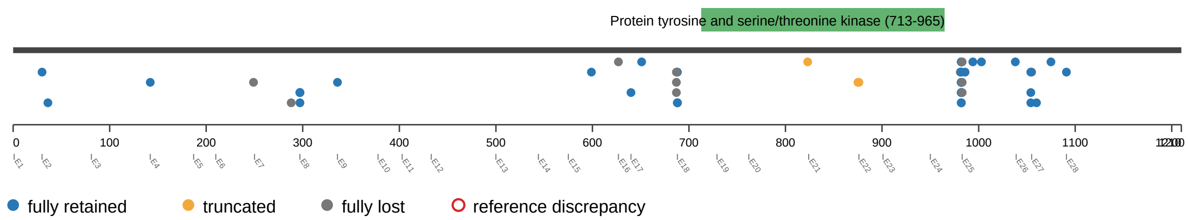
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030910-T01-IM6-551	P-0030910-T01-IM6		SEL1L-EGFR	SEL1L	in-frame	three_prime	55222485	8	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	SEL1L (NM_005065) - EGFR (NM_005228) rearrangement: t(7;14)(p11.2;q31.1)(chr7:g.55222485::chr14:g.81986576) | Protein Fusion: in frame {SEL1L:EGFR}	NA
EVT-P-0052976-T02-IM7-552	P-0052976-T02-IM7		TNS3-EGFR	TNS3	in-frame	three_prime	55222307	8	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	TNS3 (NM_022748) - EGFR (NM_005228) fusion: c.1025-2026:TNS3_c.889+462:EGFRinv | Protein Fusion: in frame {TNS3:EGFR}	NA
EVT-P-0006873-T01-IM5-554	P-0006873-T01-IM5		TUT7-EGFR	TUT7	unknown	five_prime	55238031	15	627	True	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region	Protein tyrosine and serine/threonine kinase	Growth factor receptor domain IV	EGFR (NM_005228) - ZCCHC6 (NM_024617) rearrangement : : t(7;9)(chr7p11.2;chr9q21.33)(chr7:g.55238031::chr9:g.88930920) | Protein fusion: mid-exon (EGFR-ZCCHC6)	NA
EVT-P-0015419-T01-IM6-555	P-0015419-T01-IM6		VPS41-EGFR	VPS41	unknown	five_prime	55268890	25	986	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228)- VPS41 (NM_014396) Rearrangement : c.2956:EGFR_c.247-4922:VPS41inv | Protein fusion: mid-exon (EGFR-VPS41)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030222-T01-IM6-557	P-0030222-T01-IM6		VWC2-EGFR	VWC2	out-of-frame	three_prime	55259621	22	876	True	disrupted	0.3557312252964427	True		Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase	VWC2 (NM_198570) - EGFR (NM_005228) rearrangement: c.827-37789:VWC2_c.2625+54:EGFRdel | Protein Fusion: out of frame {VWC2:EGFR}	NA
EVT-P-0019922-T01-IM6-558	P-0019922-T01-IM6		YIF1B-EGFR	YIF1B	unknown	three_prime	55238939	17	640	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EGFR (NM_005228) rearrangement: t(7;19)(p22.1;q13.2)(chr7:g.5523893::chr19:g.38806683) | Antisense Fusion	NA
EVT-P-0030222-T01-IM6-559	P-0030222-T01-IM6		ZPBP-EGFR	ZPBP	unknown	three_prime	55209996	2	36	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			ZPBP (NM_007009) - EGFR (NM_005228) rearrangement: c.487+5719:ZPBP_c.106:EGFRinv | Protein Fusion: mid-exon {ZPBP:EGFR}	NA

Figures

domain retention outliers

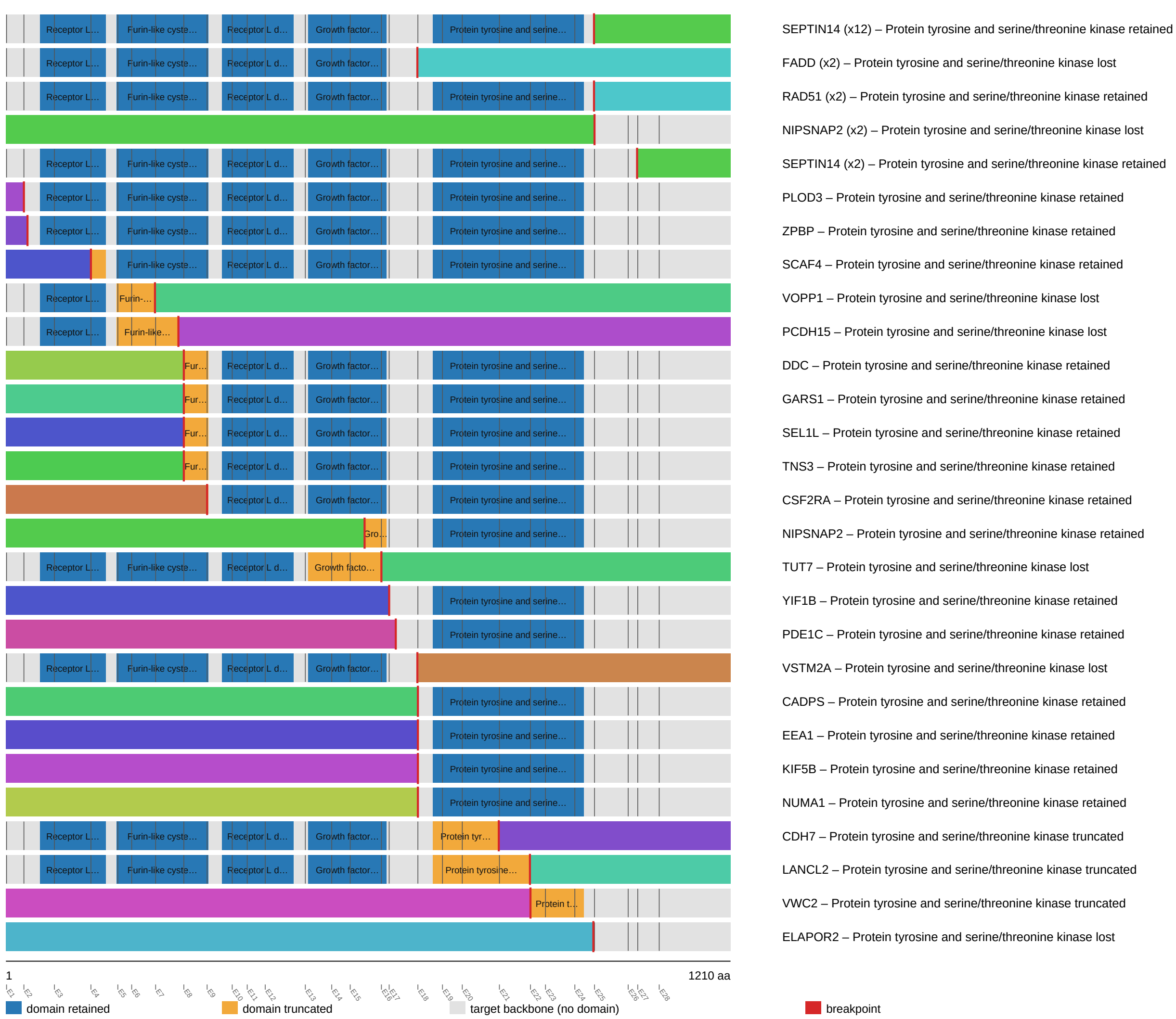
EGFR domain-retention track



fusion schematic

EGFR fusion-transcript schematic

partner block → breakpoint → retained EGFR portion (domain-colored, exon ticks)

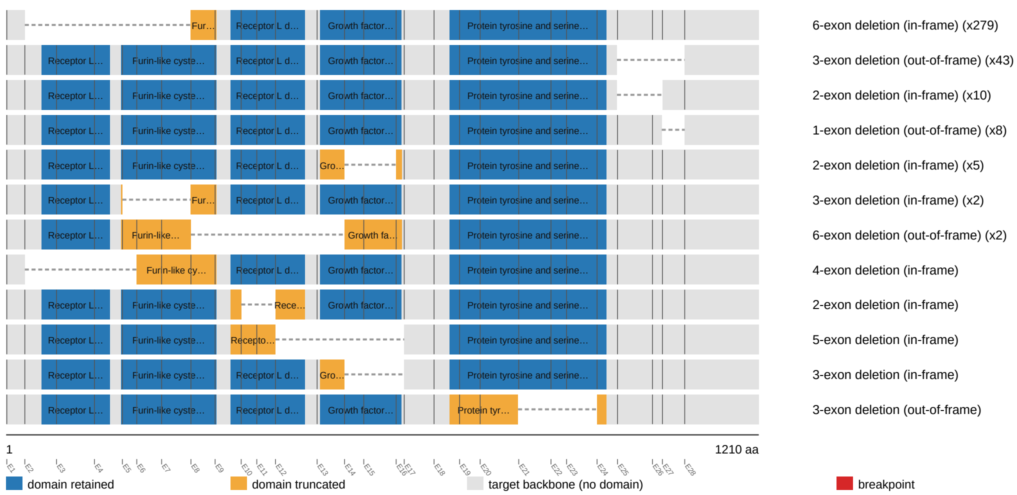


Showing the top 28 of 40 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic

EGFR intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk_impact_50k_2026